Data Analytics (DA)

Module -1 - Excel

Excel Assignment 1 – Data Exploration

1. Sum, Count, and Average

i) Sum

I6 | =SUM(D2:D35)

	A	B	C	D	E	F	G	H	I
1	Product ID	Product Name	Brand Name	Price ($)	Quantity	Category			
2	28-JAN-US	Laptop	Dell	1000	30	Electronics			
3	15-FEB-US	Sneakers	Nike	80	15	Fashion	1)	*Sum, Count, and Average*	
4	03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen			
5	11-APR-US	Smartphone	Samsung	900	25	Electronics		**Total price of all products**	(SUM)
6	22-MAY-US	Backpack	North Face	70	20	Outdoor			**10100**
7	07-JUN-UK	Headphones	Sony	200	45	Electronics		**How many products exist**	
8	19-JUL-UK	T-shirt	Adidas	30	5	Fashion			
9	23-AUG-UK	Blender	Ninja	90	35	Kitchen		**Average price of the products**	
10	05-SEP-UK	Tablet	Apple	500	50	Electronics			
11	14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor			
12	17-JUN-IN	Laptop	HP	950	25	Electronics			
13	25-NOV-AU	Sneakers	Adidas	90	40	Fashion			

ii) Count

I8 | =COUNT(D2:D35)

	A	B	C	D	E	F	G	H	I
1	Product ID	Product Name	Brand Name	Price ($)	Quantity	Category			
2	28-JAN-US	Laptop	Dell	1000	30	Electronics			
3	15-FEB-US	Sneakers	Nike	80	15	Fashion	1)	*Sum, Count, and Average*	
4	03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen			
5	11-APR-US	Smartphone	Samsung	900	25	Electronics		**Total price of all products**	(SUM)
6	22-MAY-US	Backpack	North Face	70	20	Outdoor			**10100**
7	07-JUN-UK	Headphones	Sony	200	45	Electronics		**How many products exist**	(Count)
8	19-JUL-UK	T-shirt	Adidas	30	5	Fashion			**34**
9	23-AUG-UK	Blender	Ninja	90	35	Kitchen		**Average price of the products**	
10	05-SEP-UK	Tablet	Apple	500	50	Electronics			
11	14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor			
12	17-JUN-IN	Laptop	HP	950	25	Electronics			

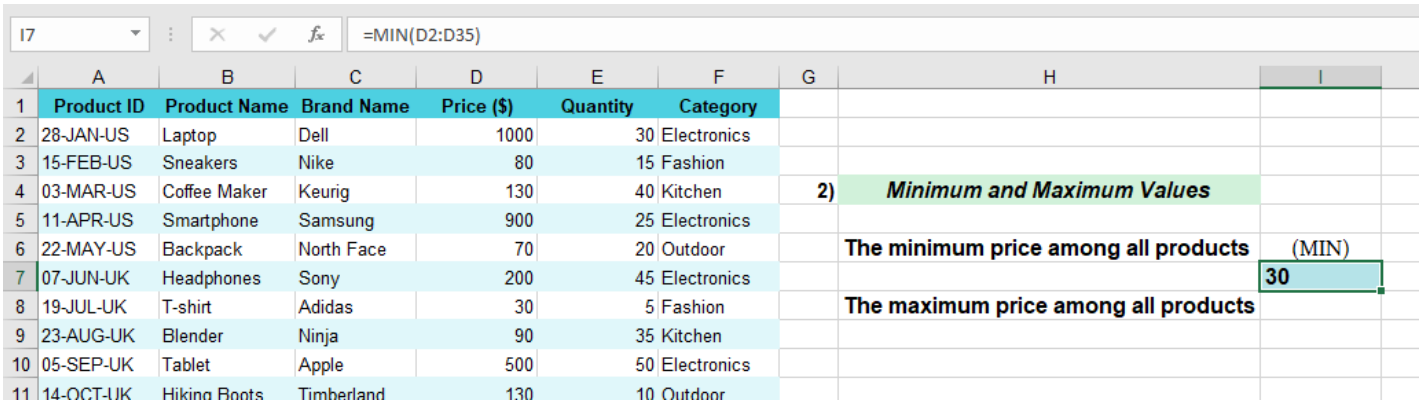
iii) Average

I10 | =AVERAGE(D2:D35)

	A	B	C	D	E	F	G	H	I
1	Product ID	Product Name	Brand Name	Price ($)	Quantity	Category			
2	28-JAN-US	Laptop	Dell	1000	30	Electronics			
3	15-FEB-US	Sneakers	Nike	80	15	Fashion	1)	*Sum, Count, and Average*	
4	03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen			
5	11-APR-US	Smartphone	Samsung	900	25	Electronics		**Total price of all products**	(SUM)
6	22-MAY-US	Backpack	North Face	70	20	Outdoor			**10100**
7	07-JUN-UK	Headphones	Sony	200	45	Electronics		**How many products exist**	(Count)
8	19-JUL-UK	T-shirt	Adidas	30	5	Fashion			**34**
9	23-AUG-UK	Blender	Ninja	90	35	Kitchen		**Average price of the products**	(Average)
10	05-SEP-UK	Tablet	Apple	500	50	Electronics			**297.05882**
11	14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor			
12	17-JUN-IN	Laptop	HP	950	25	Electronics			

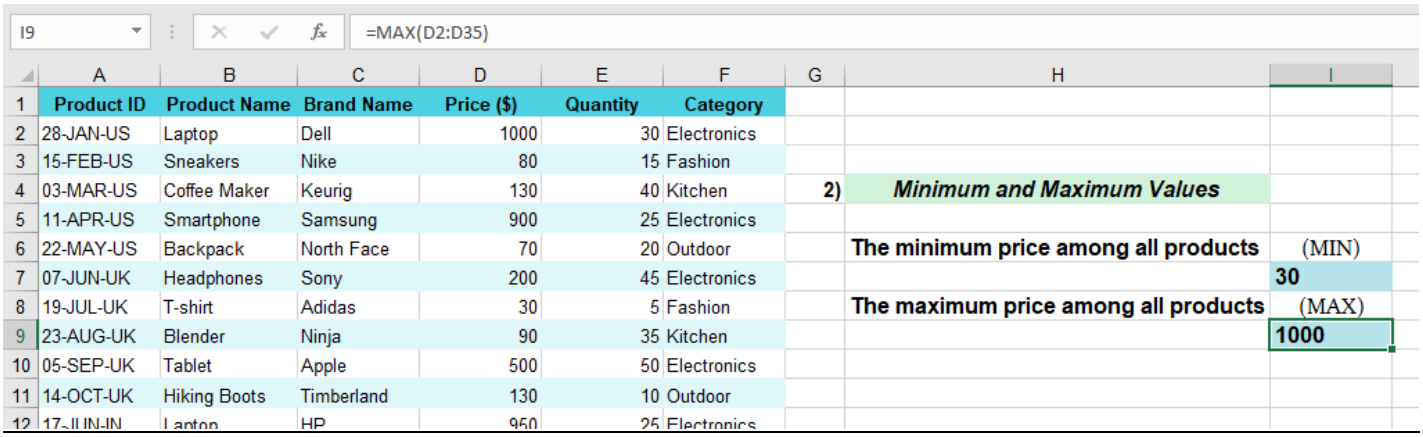
2. Minimum and Maximum Values

i) MIN



	A	B	C	D	E	F	G	H	I
1	Product ID	Product Name	Brand Name	Price ($)	Quantity	Category			
2	28-JAN-US	Laptop	Dell	1000	30	Electronics			
3	15-FEB-US	Sneakers	Nike	80	15	Fashion			
4	03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	2)	*Minimum and Maximum Values*	
5	11-APR-US	Smartphone	Samsung	900	25	Electronics			
6	22-MAY-US	Backpack	North Face	70	20	Outdoor		**The minimum price among all products**	(MIN)
7	07-JUN-UK	Headphones	Sony	200	45	Electronics			**30**
8	19-JUL-UK	T-shirt	Adidas	30	5	Fashion		**The maximum price among all products**	
9	23-AUG-UK	Blender	Ninja	90	35	Kitchen			
10	05-SEP-UK	Tablet	Apple	500	50	Electronics			
11	14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor			

ii) MAX



	A	B	C	D	E	F	G	H	I
1	Product ID	Product Name	Brand Name	Price ($)	Quantity	Category			
2	28-JAN-US	Laptop	Dell	1000	30	Electronics			
3	15-FEB-US	Sneakers	Nike	80	15	Fashion			
4	03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	2)	*Minimum and Maximum Values*	
5	11-APR-US	Smartphone	Samsung	900	25	Electronics			
6	22-MAY-US	Backpack	North Face	70	20	Outdoor		**The minimum price among all products**	(MIN)
7	07-JUN-UK	Headphones	Sony	200	45	Electronics			**30**
8	19-JUL-UK	T-shirt	Adidas	30	5	Fashion		**The maximum price among all products**	(MAX)
9	23-AUG-UK	Blender	Ninja	90	35	Kitchen			**1000**
10	05-SEP-UK	Tablet	Apple	500	50	Electronics			
11	14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor			
12	17-JUN-IN	Laptop	HP	950	25	Electronics			

3. Logical Function – IF

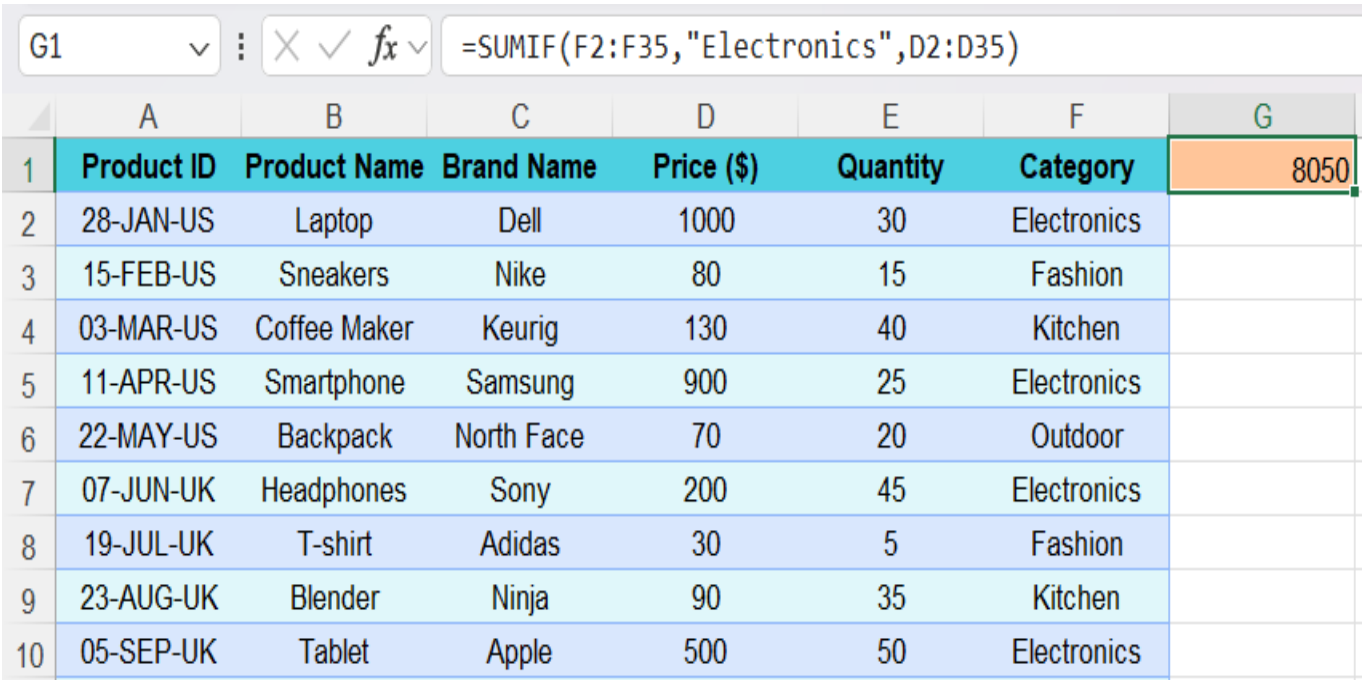
IF function



	A	B	C	D	E	F	G
1	Product ID	Product Name	Brand Name	Price ($)	Quantity	Category	Price Range
2	28-JAN-US	Laptop	Dell	1000	30	Electronics	High price
3	15-FEB-US	Sneakers	Nike	80	15	Fashion	Standard price
4	03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	Standard price
5	11-APR-US	Smartphone	Samsung	900	25	Electronics	High price
6	22-MAY-US	Backpack	North Face	70	20	Outdoor	Standard price
7	07-JUN-UK	Headphones	Sony	200	45	Electronics	Standard price
8	19-JUL-UK	T-shirt	Adidas	30	5	Fashion	Standard price
9	23-AUG-UK	Blender	Ninja	90	35	Kitchen	Standard price
10	05-SEP-UK	Tablet	Apple	500	50	Electronics	High price
11	14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	Standard price

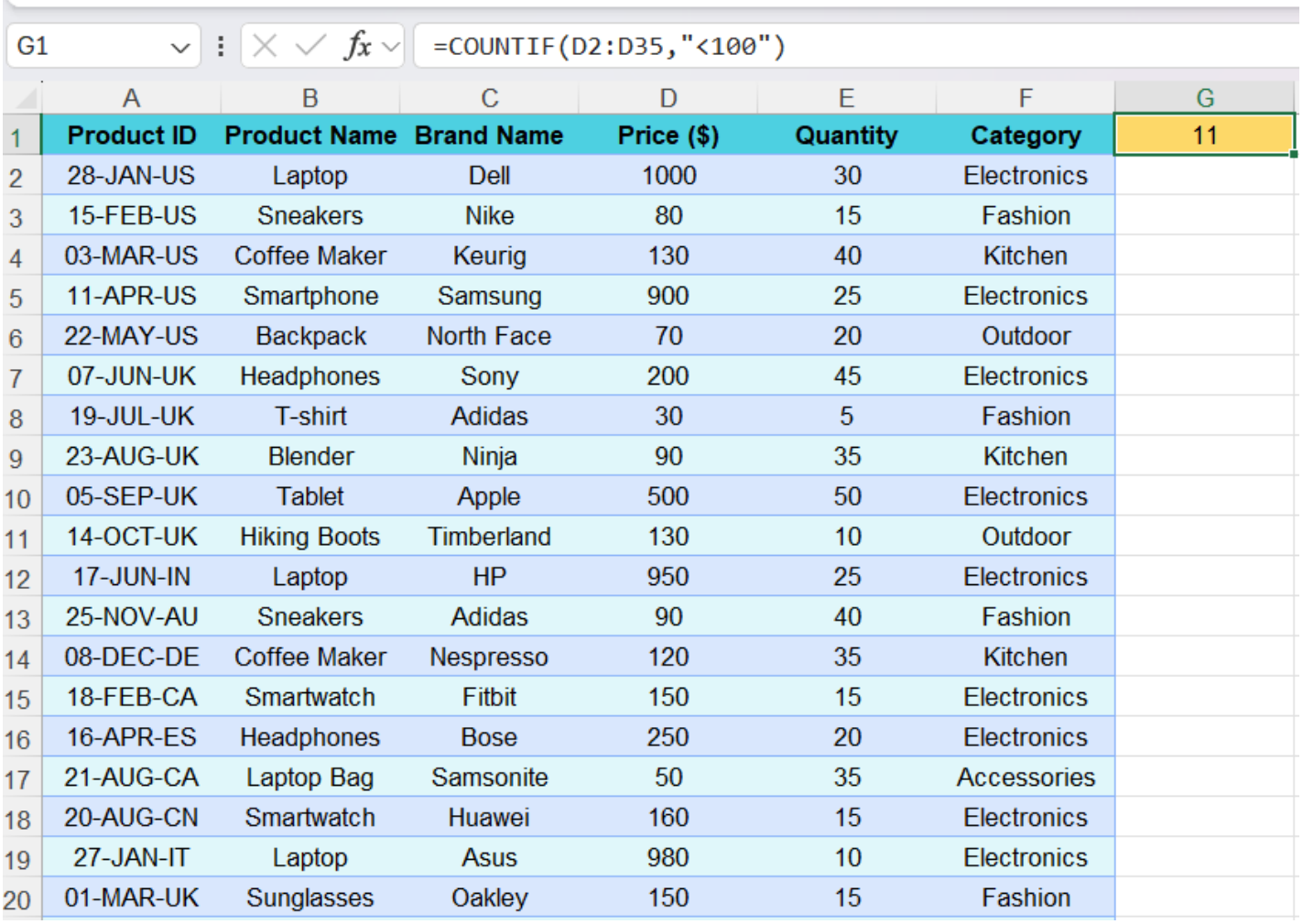
4. Conditional Functions – SUMIF and COUNTIF

SUMIF function



	A	B	C	D	E	F	G
1	Product ID	Product Name	Brand Name	Price ($)	Quantity	Category	8050
2	28-JAN-US	Laptop	Dell	1000	30	Electronics	
3	15-FEB-US	Sneakers	Nike	80	15	Fashion	
4	03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	
5	11-APR-US	Smartphone	Samsung	900	25	Electronics	
6	22-MAY-US	Backpack	North Face	70	20	Outdoor	
7	07-JUN-UK	Headphones	Sony	200	45	Electronics	
8	19-JUL-UK	T-shirt	Adidas	30	5	Fashion	
9	23-AUG-UK	Blender	Ninja	90	35	Kitchen	
10	05-SEP-UK	Tablet	Apple	500	50	Electronics	

COUNTIF function



	A	B	C	D	E	F	G
1	Product ID	Product Name	Brand Name	Price ($)	Quantity	Category	11
2	28-JAN-US	Laptop	Dell	1000	30	Electronics	
3	15-FEB-US	Sneakers	Nike	80	15	Fashion	
4	03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	
5	11-APR-US	Smartphone	Samsung	900	25	Electronics	
6	22-MAY-US	Backpack	North Face	70	20	Outdoor	
7	07-JUN-UK	Headphones	Sony	200	45	Electronics	
8	19-JUL-UK	T-shirt	Adidas	30	5	Fashion	
9	23-AUG-UK	Blender	Ninja	90	35	Kitchen	
10	05-SEP-UK	Tablet	Apple	500	50	Electronics	
11	14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	
12	17-JUN-IN	Laptop	HP	950	25	Electronics	
13	25-NOV-AU	Sneakers	Adidas	90	40	Fashion	
14	08-DEC-DE	Coffee Maker	Nespresso	120	35	Kitchen	
15	18-FEB-CA	Smartwatch	Fitbit	150	15	Electronics	
16	16-APR-ES	Headphones	Bose	250	20	Electronics	
17	21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	
18	20-AUG-CN	Smartwatch	Huawei	160	15	Electronics	
19	27-JAN-IT	Laptop	Asus	980	10	Electronics	
20	01-MAR-UK	Sunglasses	Oakley	150	15	Fashion	

5. Text Formatting – LEFT, RIGHT, MID

LEFT function

G2 =LEFT(A2,2)

	A	B	C	D	E	F	G
1	Product Il	Product Nam	Brand Nam	Price ($)	Quantity	Category	Left Function
2	28-JAN-US	Laptop	Dell	1000	30	Electronics	28
3	15-FEB-US	Sneakers	Nike	80	15	Fashion	15
4	03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	03
5	11-APR-US	Smartphone	Samsung	900	25	Electronics	11
6	22-MAY-US	Backpack	North Face	70	20	Outdoor	22
7	07-JUN-UK	Headphones	Sony	200	45	Electronics	07
8	19-JUL-UK	T-shirt	Adidas	30	5	Fashion	19
9	23-AUG-UK	Blender	Ninja	90	35	Kitchen	23
10	05-SEP-UK	Tablet	Apple	500	50	Electronics	05
11	14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	14

RIGHT function

H2 =RIGHT(A2,2)

	A	B	C	D	E	F	G	H
1	Product Il	Product Nam	Brand Nam	Price ($)	Quantity	Category	Left Function	Right Function
2	28-JAN-US	Laptop	Dell	1000	30	Electronics	28	US
3	15-FEB-US	Sneakers	Nike	80	15	Fashion	15	US
4	03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	03	US
5	11-APR-US	Smartphone	Samsung	900	25	Electronics	11	US
6	22-MAY-US	Backpack	North Face	70	20	Outdoor	22	US
7	07-JUN-UK	Headphones	Sony	200	45	Electronics	07	UK
8	19-JUL-UK	T-shirt	Adidas	30	5	Fashion	19	UK
9	23-AUG-UK	Blender	Ninja	90	35	Kitchen	23	UK
10	05-SEP-UK	Tablet	Apple	500	50	Electronics	05	UK

MID function

I2 =MID(A2,4,3)

	A	B	C	D	E	F	G	H	I
1	Product Il	Product Nam	Brand Nam	Price ($)	Quantity	Category	eft Func	Right Func	Mid Function
2	28-JAN-US	Laptop	Dell	1000	30	Electronics	28	US	JAN
3	15-FEB-US	Sneakers	Nike	80	15	Fashion	15	US	FEB
4	03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	03	US	MAR
5	11-APR-US	Smartphone	Samsung	900	25	Electronics	11	US	APR
6	22-MAY-US	Backpack	North Face	70	20	Outdoor	22	US	MAY
7	07-JUN-UK	Headphones	Sony	200	45	Electronics	07	UK	JUN
8	19-JUL-UK	T-shirt	Adidas	30	5	Fashion	19	UK	JUL
9	23-AUG-UK	Blender	Ninja	90	35	Kitchen	23	UK	AUG
10	05-SEP-UK	Tablet	Apple	500	50	Electronics	05	UK	SEP